

Name: _____

Date: _____

SOLVING LINEAR EQUATIONS, (EQUATIONS WITH UNKNOWN ON BOTH SIDES)

LO: I can solve linear equations where the unknown appears on both sides.

Unknowns on both sides

When **x appears on both sides** of the equation, collect all x terms on one side and all number terms on the other by using **inverse operations**.

Strategy: Move the smaller x term to the other side (to keep x positive), then isolate x.

Quick examples

●	$5x + 3 = 2x + 12$: subtract $2x$, get $3x + 3 = 12$, $x = 3$	collect x terms
●	$7x - 4 = 3x + 8$: subtract $3x$, get $4x - 4 = 8$, $x = 3$	move smaller x
●	Always move the smaller x coefficient to keep x positive	good practice
●	$5x + 3 = 2x + 12$ gives $7x = 15$	cannot add x terms across =

Key idea

Collect x terms on one side, number terms on the other, then solve.

$$\boxed{5x} + 3 = \boxed{2x} + 12 \quad 3x = 9 \quad x = 3$$

Subtract $2x$ from both sides, then subtract 3 from both sides

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - x terms positive

Solve $5x + 2 = 3x + 10$

Subtract $3x$: $2x + 2 = 10$

Subtract 2: $2x = 8$

Divide by 2: $x = 4$

$$x = 4$$

Move the smaller x term ($3x$) to get a positive coefficient.

Example 2 - larger coefficient on right

Solve $2x + 9 = 5x - 3$

Subtract $2x$: $9 = 3x - 3$

Add 3: $12 = 3x$

Divide by 3: $x = 4$

$$x = 4$$

Move $2x$ to keep x positive on the right.

Example 3 - negative terms

Solve $6x - 5 = 4x + 7$

Subtract $4x$: $2x - 5 = 7$

Add 5: $2x = 12$

Divide by 2: $x = 6$

$$x = 6$$

Subtract the smaller x coefficient from both sides.

Example 4 - with brackets

Solve $3(x + 2) = x + 14$

Expand: $3x + 6 = x + 14$

Subtract x: $2x + 6 = 14$

Subtract 6: $2x = 8$, $x = 4$

$$x = 4$$

Expand brackets first, then collect like terms.

YOUR TURN – PRACTICE (deliberate practice)

1. Simple equations

Solve each equation.

a) $4x + 1 = 2x + 7$

b) $6x + 3 = 4x + 11$

c) $7x - 2 = 3x + 10$

d) $5x + 4 = x + 20$

2. Larger coefficient on right

Move x to the side with the larger coefficient.

a) $2x + 10 = 5x + 1$

b) $x + 9 = 4x$

c) $3x + 1 = 7x - 11$

d) $x + 14 = 3x + 2$

3. With negatives

Solve carefully.

a) $8x - 3 = 5x + 9$

b) $6x - 7 = 2x + 5$

c) $9x - 10 = 4x + 5$

d) $3x - 8 = x - 2$

4. With brackets

Expand, then solve.

a) $2(x + 5) = x + 13$

b) $4(x + 1) = 2x + 10$

c) $3(x + 2) = 2(x + 4)$

d) $5(x - 1) = 3(x + 1)$

5. Mixed problems

Solve each equation.

a) $10x + 3 = 7x + 12$

b) $2(3x + 4) = 4(x + 5)$

c) $9 - 2x = x - 6$

d) $4x + 7 = 19 - 2x$

e) $3(2x - 1) = 2(x + 3) + 7$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $5x + 3 = 2x + 12$ becomes $3x + 3 = 12$ (subtract $2x$ from both sides) ☐

Correct answer: _____

Reason: _____

b) $5x + 3 = 2x + 12$ becomes $7x + 3 = 12$ ☐

Correct answer: _____

Reason: _____

c) Move the smaller x term to avoid negative coefficients ☐

Correct answer: _____

Reason: _____

d) You can add x terms from different sides: $5x + 2x = 7x$ ☐

Correct answer: _____

Reason: _____

e) After collecting terms, solve as a one-step or two-step equation ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Two taxi companies: Company A charges **£3 + £2 per mile**. Company B charges **£5 + £1.50 per mile**. For what distance are they the same price? Which is cheaper for a 10-mile journey?

Expression: _____

Simplified: _____

b) A triangle has sides **($3x + 1$) cm**, **($2x + 5$) cm**, and **($4x - 2$) cm**. An equilateral triangle has perimeter equal to this triangle and side length **($2x + 3$) cm**. Find x and the perimeters.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

